T03 Y-Sorting

CRE132 worksheet

What you'll make

In T01 you could walk "behind" a tree and still be drawn on top of its leaves. Here you'll fix that, so that above a trunk you go behind the tree and below it you're in front. This is called **Y-sorting**: things lower on the screen are drawn in front.

There's no code in this sample, only settings. By the end you'll know the three things Y-sorting needs (the same sorting layer, a sort axis, and pivots at the feet), what a **Sorting Group** does, and how a **prefab variant** changes one setting without copying a whole prefab.



Figure 1: The finished grove: rows of trees close together with a path between them.

Before you start

1. Unzip `CRE132_T03_YSorting_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_T03_YSorting_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/TopDown/T03_YSorting`.
3. Press **Play** and walk around the trees (WASD or arrows). You can't walk through the trunks, but which of you is drawn on top has nothing to do with where you stand: walking past a tree doesn't swap you from behind it to in front of it. Press Play again to stop.

The scene is a grove of **Tree (Y-Sorted)** trees, a few rocks and bushes, and the **Player** (T01's movement and T02's animation). Select the **Player** in the Hierarchy: it has a child called **Sprite**. Keep that in mind; part 4 explains why.

1. Why sorting layers come first

In T01 the trees were on the **Objects** sorting layer and the player on **Characters**, which comes after Objects in the list (**Edit > Project Settings > Tags and Layers > Sorting Layers**). So the player was always drawn on top, whatever its position.

Things that should overlap correctly must share a sorting layer (and **Order in Layer**). Select a tree here and look at its **Sprite Renderer**: its **Sorting Layer** is **Characters**, the same as the player's **Sprite**.

2. Same layer: a prefab variant

The trees didn't change layer by editing T01's **Tree (Top-Down)** prefab, or every T01 tree would have changed too. They're a **prefab variant**: a copy that keeps everything from the original except the settings you change.

1. In the Project window, open `Assets/_game/Prefabs/Environment` and double-click **Tree (Y-Sorted)** to open it.
2. Look at its **Sprite Renderer**: **Sorting Layer** (**Characters**) and **Sprite Sort Point** (**Pivot**, part 4 explains it) are in **bold** with a blue line in the margin. Those are the variant's overrides; everything else comes from Tree (Top-Down). Click the back arrow at the top left of the Hierarchy to leave the prefab.

The rocks are still T01's **Rock (Top-Down)**, on **Objects**. Make a variant for them:

1. In `Assets/_game/Prefabs/Environment`, right-click **Rock (Top-Down)** and choose **Create > Prefab Variant**. Name it **My Rock (Y-Sorted)**.
2. Double-click it to open it. On its **Sprite Renderer**, set **Sorting Layer** to **Characters** and **Sprite Sort Point** to **Pivot**. Both turn bold. Click the back arrow to leave the prefab.
3. Drag two **My Rock (Y-Sorted)** into the scene, near a path where you can walk round them.

Test it: press Play. Nothing sorts by position yet, for the trees or your rocks. The next part switches that on.

3. Sort by Y: the 2D Renderer

Sharing a layer isn't enough: Unity also needs to know *which way* to sort. That's set on the project's 2D Renderer.

1. In the Project window, open `Assets/Settings` and click **Renderer2D**.
2. In the Inspector, set **Transparency Sort Mode** to **Custom Axis**.
3. A **Transparency Sort Axis** field appears below it. Set it to X **0**, Y **1**, Z **0**.

This tells Unity: on the same layer and order, draw things with a lower Y in front.

- **It's a project-wide setting**, so every scene in the project now sorts this way.
- **Default** sorts by distance from the camera. In 2D everything is at the same distance, so the order has nothing to do with where things stand. That's what you saw before you started.

Test it: press Play and walk round a tree. Above the trunk you now go behind the leaves. But stand just *below* a trunk: you're still drawn behind it, although you're in front. Part 5 fixes that.

4. Sorting by the feet

Unity has to pick *one point* of each sprite to compare. By default a sprite sorts by its centre, which for a tall tree is somewhere in the leaves.

- **The tree's pivot is at the bottom of its trunk.** Select `objects` in `Assets/_game/Art/Sprites/MysticWoods`, click **Open Sprite Editor** and select the tree: its **Pivot** is **Bottom Center**. With **Sprite**

Sort Point set to **Pivot**, the tree sorts by where it touches the ground. The rock's pivot is at its bottom too.
- **The player's frames are 48 pixels square with the pivot in the middle**, but the feet are 19 pixels lower. Instead of changing every frame's pivot, the player is built in two parts. Select the **Player** (the root): it stands at the feet, with the **Rigidbody 2D**, the **Box Collider 2D** at the feet (**Offset** 0, 0.15) and **Top Down Movement**. Now select its child **Sprite**: it holds the **Sprite Renderer**, the **Animator** and **Directional Animator**, and its **Position** is 0, 1.1875, 0. That moves the picture up 19 pixels: 19 / 16 pixels per unit is 1.1875 units.

That's why sorting is still off: the **Sprite** child sorts by its own centre, which is 1.1875 units above your feet.

5. A Sorting Group

Select the **Player** (the root, not **Sprite**). Click **Add Component** and choose **Sorting Group**. Set its **Sorting Layer** to **Characters**.

- **A Sorting Group sorts everything under it as one**, using the root's position, which is the feet.
- **It keeps future parts together.** A sword, a shadow or a hat added as more children will always be drawn with the player, never with a tree in between.

Test it: press Play. Walk just above and just below each trunk and watch the swap happen right at the feet. Do the same round your rocks.

Check your work

- ☐ **My Rock (Y-Sorted)** is a prefab variant on **Characters** with **Sprite Sort Point Pivot**.
- ☐ **Renderer2D** is on **Custom Axis**, with the axis (0, 1, 0).
- ☐ The **Player** root has a **Sorting Group** on **Characters**.
- ☐ Above a trunk the leaves cover the player; below a trunk the player is in front, and the swap happens at the feet. Your rocks swap the same way.
- ☐ The player still can't walk through trunks.
- ☐ Compare with `CRE132_T03_YSorting_Finished` if something behaves differently.

Key points

- **Y-sorting needs three things:** the same sorting layer, **Custom Axis (0, 1, 0)**, and pivots (or roots) at the feet.
- **A Sorting Group** sorts a character made of several sprites as one.
- **Prefab variants** change one setting without copying the whole prefab.

If something goes wrong

- **Which one is on top doesn't change as you walk: Renderer2D**'s **Transparency Sort Mode** is still **Default**, or the axis isn't (0, 1, 0).
- **The player is always on top of a rock:** that rock is still a **Rock (Top-Down)** on **Objects**. Check you placed **My Rock (Y-Sorted)**.
- **The swap happens too high, in the middle of the player:** the **Sorting Group** is missing from the Player root, or you added it to the **Sprite** child instead.
- **A rock swaps at its middle, not its bottom:** its **Sprite Sort Point** is still **Center**. Open **My Rock (Y-Sorted)** and set it to **Pivot**.
- **The player is drawn behind everything, or in front of everything:** the Sorting Group's **Sorting Layer** isn't **Characters**.

Try it

1. Make the bushes Y-sorted too: a variant of **Bush (Top-Down)** on **Characters**, with **Sprite Sort Point Pivot**.
2. Set your rock variant's **Sprite Sort Point** back to **Center** and find where sorting goes wrong.
3. Add a shadow under the player: another child of **Player** with its own Sprite Renderer, placed at the feet. Thanks to the Sorting Group it always stays with the player.